# TASEHINTA

## Know when the imbalance price will break away from spot, and why, before it happens.

A 10-hour imbalance-price forecast and situational view for Turku Energia, quarter-hour by quarter-hour.

**36.7 % better than the baseline**

Team EDISS #1 · Since AI 2026 · Turku Energia challenge





THE PROBLEM

# Every missed plan is settled at a price nobody knows yet

Finland's imbalance price is set every 15 minutes. It follows spot until the grid runs short of power or into a surplus, then it can jump far either way.

**4,000 €/MWh** — the imbalance price on 9 Sep at 20:30, with spot at 200

**1 in 4** — quarter-hours in September ended more than 50 €/MWh from spot

**15 min** — settlement period: every quarter-hour is its own price

Source: organisers' validation data, 7 Sep – 4 Oct 2026

OUR SOLUTION

# Every hour, the next 40 quarter-hours, four answers each

### What will it cost?

A median imbalance price and an honest 80 % range around it

### Which way is the grid going?

The probability of up-regulation, down-regulation or none

### Will it spike?

The chance of a price more than 50 €/MWh above or below spot

### Why?

The few factors that move the forecast most, strongest first

**Then it shows only what deserves attention** and says what to check, never what to trade: open data can't see Turku Energia's own position.

# One model, built only from what was known at the time



## Published data

Only what was out before the hour: market, balancing and grid data, Fingrid real-time, weather runs, 19 months of history



## Features

For each of the 40 quarter-hours: recent state, real-time grid, wind-forecast revisions, the shape of the day-ahead price



## One ensemble model

Two gradient-boosting components, averaged: their errors partly cancel



## Forecast + dashboard

The submitted CSV and the situational view

**Forecast the gap, not the price** Spot is known the day before; the unknown is how far the imbalance price lands from it.

**Direction first** Short of power: above spot. Surplus: below. Price, range, direction and spike risk tell one story.

**History made comparable** Pre-June prices rebuilt under the June 2026 rule: 99.5 % exact where it can be checked.

**Nothing from the future** Every input used only from its publication time, enforced by automated leak tests.

# More accurate than the baseline on every score

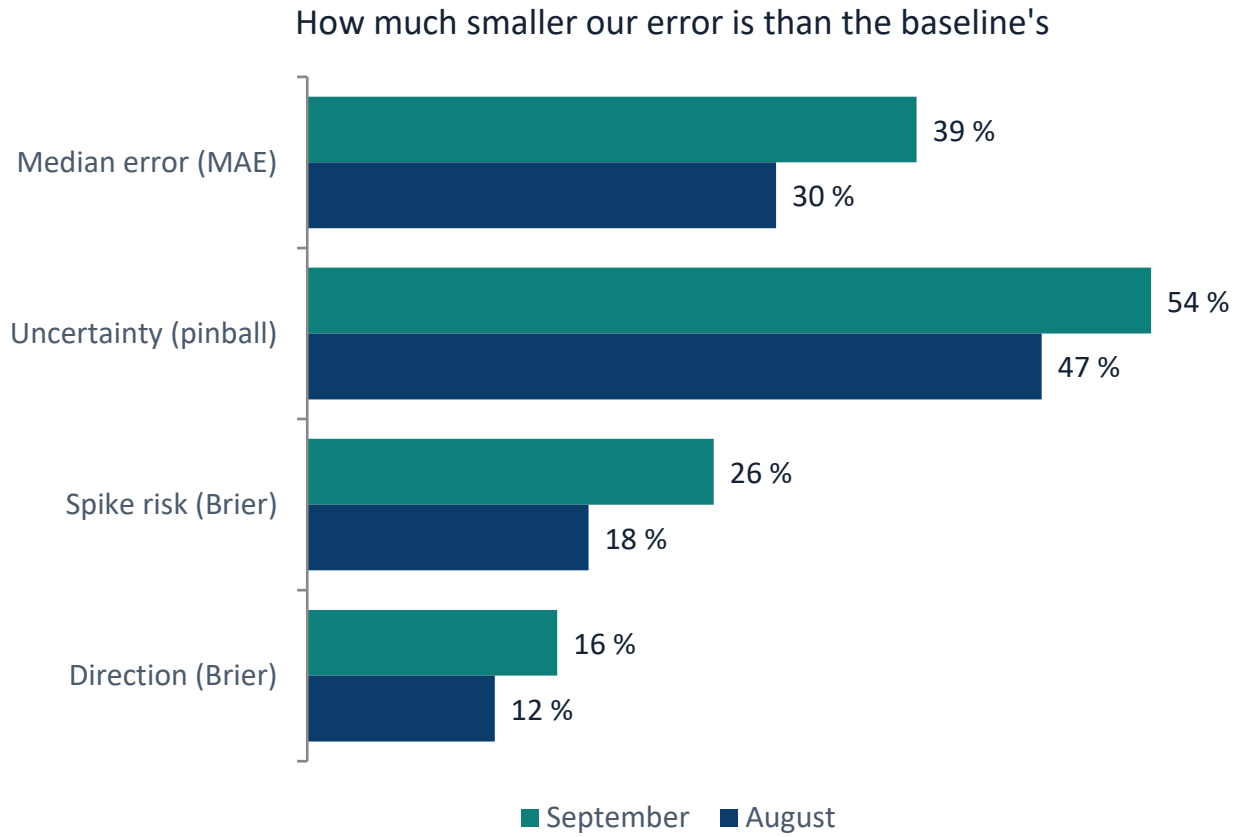
**36.7 %**

better than the organisers' baseline on their validation month (Sep), weighted by the challenge's points

**29.0 %**

on August, a second month the model never saw

40.2 vs 66.2 €/MWh median error · better in every September week




Scored with the organisers' own score.py; each month forecast using only data published before it.

# See what's worth checking, and replay any past hour




## Only what matters

Spike risk up, regulation expected, forecast moved, an uncertain signal

## One time axis

Price and range, direction and spike risk on one 15-minute grid

## Move through time

Scroll back in 15-minute steps; nothing published later is shown

Live: [tasehinta.lovable.app](tasehinta.lovable.app)

# From “check this” to “offer this”

### Turku Energia's own data

Plant capacity, marginal costs and the market position turn a warning into a concrete capacity suggestion

### Winter weather history

The history covers winter 2025–26, but weather-forecast runs only from April 2026; adding winter runs is the next model step

### Live operation

Run the same pipeline every hour on live Fingrid and weather data, behind an API

TASEHINTA

36.7 % better than the baseline. Calibrated uncertainty. Only data that existed at the time.

**tasehinta.lovable.app**

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